

Intrinsic explainability (white-box models)

This approach focuses on using inherently interpretable model architectures. These ‘white-box’ models are designed from the ground up to be transparent in their decision-making process. Examples include:

- **Decision trees:** These models represent decisions as a series of hierarchical rules, making the path to a prediction easily traceable.
- **Linear regression:** The relationship between input features and the output is clearly defined by coefficients, indicating the direction and magnitude of each feature’s influence.
- **Rule-based systems:** Decisions are made based on a set of explicit ‘if-then’ rules that are directly understandable by humans.

While these models offer high interpretability, they often come with limitations in terms of the complexity of the patterns they can learn and their predictive power compared to black-box models.

Post-hoc explainability (black-box models)

These methods illuminate ‘the black box’ without modifying its internal workings. Some important post-hoc methods include:

- **Feature importance:** These methods identify which input features had the most significant influence on a model’s prediction. Techniques like SHAP (SHapley Additive exPlanations) and permutation importance fall into this category.
- **Saliency maps:** Primarily used in computer vision, these techniques highlight the regions in an input image that were most important for the model’s classification decision.
- **Local Interpretable Model-agnostic Explanations (LIME):** LIME approximates the decision boundary of a complex model locally around a specific instance by training a simpler, interpretable model on perturbed versions of that instance.
- **Counterfactual explanations:** These explanations identify the minimal changes to an input that would lead to a different prediction, helping users understand ‘what if’ scenarios.
- **Attention mechanisms:** In models like transformers used in natural language processing, attention weights can provide insights into which parts of the input sequence the model focused on when making a prediction.

Selecting the appropriate explainability technique depends on the specific use case and data modality. Table 1 maps common use cases to recommended XAI methods along with their rationale.

Table 1: Mapping common use cases to recommended XAI methods

Use case	Recommended technique	Why?
Global interpretability	SHAP (SHapley Additive exPlanations)	Offers additive, consistent global explanations
Local interpretability	LIME (Local Interpretable Model-agnostic Explanations)	Explains individual predictions using local linearity
Image feature attribution	Saliency maps	Highlights important pixels influencing decision
Tabular data with correlated features	SHAP	Handles feature interaction better than LIME

Table 2: Comparison of intrinsic vs post-hoc explainability

Aspect	Intrinsic explainability	Post-hoc explainability
Definition	Built into the model architecture itself	Interpretation applied after training
Model type	Transparent (e.g., decision trees, linear models)	Black-box (e.g., neural networks, ensemble models)
Interpretability level	High, as the logic is human-understandable	Varies, depends on technique used
Performance trade-off	May sacrifice performance for interpretability	Typically maintains model performance
Example	Logistic regression showing feature coefficients	SHAP explaining a CNN’s image classification

XAI promotes ethical use and responsibility in AI applications. As agentic AI grows, XAI will become even more important. Since these systems make decisions based on complex learning and interactions, understanding their reasoning is crucial. The key objectives of XAI are shown in Figure 1.

The challenge of explaining agent autonomy

The shift from traditional AI to agentic AI adds complexity to explainability.

Autonomous agents are computer programs that can make decisions and act on their own without overt human intervention. With more and more sophisticated and pervasive agents—self-driving vehicles, trading algorithms, and medical diagnoses—the need to explain and understand